

Muhammad Adeel

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Work Experience

Positive Solutions

April 2026 - July 2026

Data Analyst

Wah Cantt, Pakistan

- Analyzed and processed large datasets using **Excel, MySQL, and Python** to support data-driven business decisions across departments.
- Built and maintained **interactive dashboards and reports** to track key performance metrics and monitor business trends over time.
- Wrote and optimized **complex SQL queries** (joins, aggregations, subqueries) to extract, clean, and organize data from company databases, improving reporting accuracy and turnaround time.
- Automated repetitive **data-processing workflows** using Python and AI tools, streamlining analysis and enabling faster business decision-making.

NAVTTC

Oct 2025 - March 2026

Python Development Trainee

Islamabad, Pakistan

- Assisted in the development and testing of **machine learning models**, contributing to data preparation and evaluation stages.
- Worked extensively with **NumPy and Pandas** for data manipulation, cleaning, and analysis.
- Participated in **data preprocessing and visualization** tasks to support model training and performance assessment.
- Gained hands-on exposure to **end-to-end AI/ML workflows** while collaborating with a team on real project tasks.

Projects

Spine Disease Detection System | CNN (EfficientNetV2S)

[Github Link](#)

- Developed a **computer vision-based medical imaging system** to detect and classify spinal disorders (Neural Foraminal Narrowing, Subarticular Stenosis, Spinal Canal Stenosis) using the RSNA Lumbar Spine dataset.
- Fine-tuned an **EfficientNetV2S** architecture achieving **95% classification accuracy** across disorder categories, with image preprocessing and augmentation to improve generalization on DICOM scans.

Third Eye - AI-Powered Assistant for Visually Impaired Persons

[Github Link](#)

- Built a real-time multi-module assistive system combining **scene description (BLIP)**, **object detection (YOLO11)**, **text extraction (EasyOCR)**, and Pakistani currency recognition using a fine-tuned **DenseNet121** model.
- Integrated **text-to-speech (gTTS)** to convert combined analysis into real-time spoken feedback, helping visually impaired users understand surroundings, read text, and identify currency notes.

Heart Disease Prediction | Ensemble ML Web App

[Github Link](#)

- Built a web app predicting heart disease likelihood on the Kaggle Playground Series S6E2 dataset via EDA, correlation analysis, and an **ensemble of XGBoost, LightGBM, and CatBoost** models.
- Achieved an **ensemble AUC of ~0.956** on binary classification of disease presence/absence using averaged model probabilities.

Movie Recommender System | Streamlit

[Github Link](#)

- Developed a **content-based movie recommendation web app** using Streamlit, suggesting similar movies based on genre, cast, and plot features for a user-searched title.

Multimodal Parkinson's Disease Detection Using Voice (Research - In Progress)

[Github Link](#)

- Designing a **multimodal cross-attention framework** combining patient audio and speech transcripts, using a **LoRA fine-tuned Wav2Vec2** model to detect Parkinson's disease while reducing memory overhead and catastrophic forgetting.
- Applying **GroupKFold cross-validation** to prevent data leakage and **SHAP-based interpretability** to explain model predictions for clinical trust.

Skills and Certifications

Programming Languages: Python, SQL

Libraries and Frameworks: NumPy, Pandas, PyTorch, TensorFlow, Scikit-learn, OpenCV, Flask, Streamlit

Machine Learning: Deep Learning, Computer Vision, NLP, Model Fine-Tuning (LoRA), Ensemble Methods, Model Evaluation

Data and Tools: Excel, Power BI, MySQL, Git, Jupyter

Certifications: Machine Learning Specialization - Andrew Ng (Coursera), Deep Learning Specialization - Andrew Ng (Coursera), Python Data Structures - University of Michigan (Coursera)

Education

University of Haripur

Sep 2021 - June 2025

B.Tech, Computer Science (CS)

Grade: A

FG Science Degree College, Wah Cantt

2019 - 2021

Faculty of Computer Science

Achievements

- Secured the **Top 5th Best Final Year Project** award from the Department of Information Technology, University of Haripur, for the Spine Disease Detection System.
- Achieved **95% classification accuracy** while fine-tuning an EfficientNetV2S model for multi-class spinal disorder detection on real medical imaging data.
- Participated in the **Machine Learning Bootcamp** and **The Robotics with AI Seminar** at University of Haripur, strengthening practical AI/ML expertise.